CIVIL ENGINEERING & STRUCTURAL DESIGN

TASK 4: MINI PROJECT

CodeAlpha Internship

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Task	Structural Design Proposal — conceptual pedestrian truss bridge
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Project title: Conceptual design of a small pedestrian truss bridge

1. Executive summary

This mini project proposes a small pedestrian bridge to provide a safe crossing over a narrow drainage channel or landscaped watercourse. A steel truss concept is selected because triangular members can create a stiff and efficient load-carrying system. The proposal covers the concept, preliminary dimensions, load categories, analysis approach, sustainability, construction sequence, and safety considerations. This is a planning-level academic proposal, not a final design.

2. Project objectives

- Provide a clear pedestrian route across a small obstacle.
- Use a simple structural form with an understandable load path.
- Consider dead load, pedestrian live load, wind, lateral stability, and foundation conditions.
- Encourage durable materials, efficient fabrication, safe construction, and low-maintenance operation.

3. Proposed concept and preliminary geometry

The conceptual bridge has a 12 m span and a 2 m clear walking width. Two side trusses support transverse floor beams and a deck. Handrails and edge protection are included. The deck should use a durable anti-slip surface selected for the site and exposure conditions. Final dimensions must be checked against required clearance, accessibility needs, hydrology, and local regulations.

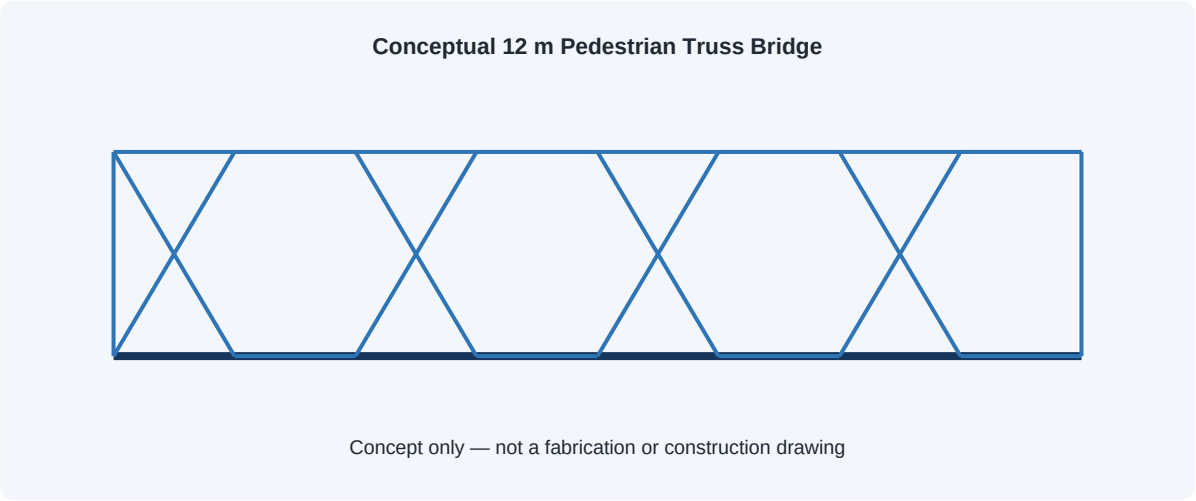

Figure 1. Indicative side elevation; this is a concept sketch, not a fabrication drawing.

4. Preliminary design assumptions

Item	Concept assumption	To confirm later
Span	12 m simply supported	Survey, levels and clearances
Walking width	2.0 m clear width	Accessibility and user demand
Main structure	Two steel side trusses	Member forces and buckling
Deck system	Floor beams and deck panels	Slip resistance, drainage, vibration
Supports	Abutments at both ends	Soil bearing, settlement and scour

5. Loads to be considered

The loads below must be identified before detailed design. Numerical values must be selected from the governing local code and project brief; this proposal does not invent site-specific code loads.

- **Dead load:** self-weight of trusses, floor beams, deck, handrails, bolts, coatings, and permanent fixtures.
- **Pedestrian live load:** people and permitted maintenance equipment. Use the code-prescribed pedestrian load and any required concentrated-load case.
- **Wind load:** pressure on trusses, handrails, and deck, including lateral forces and uplift where required.
- **Dynamic effects:** walking-induced vibration and crowd movement; check comfort as well as strength.
- **Temperature and movement:** expansion/contraction and bearing movement where relevant.
- **Seismic or accidental actions:** include where required by the site's hazard level, code, and bridge importance.
- **Water and ground effects:** flood level, debris impact, erosion, scour, and soil movement if near a channel.

6. Structural load path

Pedestrian load acts on the deck and is transferred to transverse floor beams. Floor beams transfer reactions to the two side trusses. Truss members carry axial tension or compression and transfer forces to end bearings and abutments. The abutments then transfer forces to the ground through foundations. Wind and lateral loads are resisted by bracing and the foundations.

7. Analysis and design workflow

Step 1 — Site information: obtain a topographic survey, channel width and flood levels, geotechnical report, access routes, utilities, and environmental restrictions.

Step 2 — Design criteria: identify the applicable code, load factors and combinations, material grades, serviceability limits, durability class, and design life.

Step 3 — Structural model: represent truss joints and members, deck beams, supports, and bracing. Confirm whether joints may be idealized as pinned and whether secondary bending must be included.

Step 4 — Analyze load cases: calculate member axial forces, beam bending and shear, support reactions, deflections, and vibration response.

Step 5 — Verify members and connections: check tension, compression buckling, combined forces, bolts/welds, gusset plates, fatigue where applicable, and corrosion protection.

Step 6 — Check serviceability and safety: vertical/lateral deflection, pedestrian comfort, handrail loading, deck slip resistance, drainage, access, and inspection provisions.

8. Illustrative preliminary calculation framework

For a preliminary study, let the total design vertical load per unit length be q kN/m after selecting the appropriate code loads and combinations. For a simply supported 12 m span under a uniformly distributed load, the beam-analogy support reaction is:

$$R = qL/2 = q(12)/2 = 6q \text{ kN at each end}$$

The corresponding idealized global bending moment using a simple beam analogy is:

$$M_{max} = qL^2/8 = q(12^2)/8 = 18q \text{ kN·m}$$

These expressions are only a preliminary global analogy. A truss must be designed using member forces from a structural analysis model. Final numerical design cannot be completed until code loads, self-weight, geometry, bracing, and support conditions are established.

9. Member and connection checks

- Tension members: gross-section yielding, net-section fracture, connection eccentricity, and block shear where relevant.
- Compression members: effective length, slenderness, buckling resistance, and restraint at panel points.
- Floor beams: bending, shear, local buckling, deflection, and deck attachment.
- Connections: bolts/welds and gusset plates for actual force directions, suitable edge distances, and corrosion protection.
- Lateral stability: cross-bracing and adequate resistance to wind and construction-stage loads.
- Foundations/abutments: bearing, sliding, overturning, settlement, erosion/scour, and water effects.

10. Materials and durability

Structural steel offers a high strength-to-weight ratio and can be fabricated off-site. Select a recognized steel grade after confirming local availability and code requirements. Protect steel with a coating or galvanizing system suited to the environment, and avoid details that trap water. Concrete abutments may be suitable if supported by geotechnical findings. Material selection should consider transport, maintenance, repairability, and recycling.

11. Sustainability considerations

- Use efficient member layouts to reduce unnecessary material while meeting safety requirements.
- Prefer locally available certified materials and fabrication where quality can be verified.
- Use good drainage and protective coatings to extend service life and reduce maintenance.
- Minimize disturbance to vegetation, waterways, and nearby access during construction.
- Plan for inspection, replacement of wearing surfaces, and recycling steel at end of service life.

12. Proposed construction sequence

1. Confirm survey, permits, environmental constraints, geotechnical conditions, and flood clearance.
2. Complete detailed calculations and drawings, then obtain required approvals and independent review.
3. Construct and inspect foundations and abutments after confirming bearing strata and water-management measures.
4. Fabricate truss panels and connections in a controlled workshop; check dimensions, welds/bolts, and protective coating.
5. Transport components safely and erect them using an engineered lifting and temporary-stability plan.
6. Install floor beams, deck, bracing, handrails, drainage, and anti-slip finish.
7. Complete inspection and any legally required acceptance testing before opening the bridge.

13. Risks and mitigation

Risk	Possible effect	Mitigation
Flood/scour	Support loss or approach damage	Hydrology review, suitable foundations, scour protection
Temporary instability	Erection accident or damage	Engineered sequence and temporary bracing
Corrosion	Reduced capacity over time	Coating, drainage, inspection and maintenance
Slippery deck	Pedestrian injury	Anti-slip surface, handrails, lighting and access
Wrong assumptions	Unsafe or uneconomical bridge	Survey, code checks and engineering review

14. Conclusion

A small pedestrian truss bridge is a reasonable conceptual solution for a short crossing because its triangulated form provides a clear load path and allows off-site fabrication. Before construction, the concept must become a code-compliant design using site-specific loads, a verified structural model, member and connection checks, foundation design, and professional review. Sustainability and inspection should be planned from the beginning.

15. References and design basis

- Applicable local building/bridge regulations and authority requirements (identify for the project location).
- Structural steel, concrete, and foundation standards adopted by the approving authority.
- Project-specific survey, geotechnical investigation, hydrology/flood information, and material certificates.
- Basic structural mechanics principles for trusses and simply supported beams.

Submission note

This is an educational internship proposal. Replace the assumptions with supervisor-provided project data if available. The conceptual bridge is not a certified or construction-ready design.